

8088776DBMS: LAB 3

DDL (Data Definition Language) Commands

University Fest Management System

OBJECTIVE: To learn and understand DDL statements while executing queries.

DDL (Data Definition Language) queries in MySQL are used to define and manage the structure and characteristics of your database and its objects.

Some of the important DDL commands are:

- **CREATE:** This DDL query establishes a new database object, such as a table or an index, with defined attributes and structure.
- **ALTER:** The ALTER query modifies the structure of an existing database object, enabling tasks like adding or modifying columns.
- **TRUNCATE:** This DDL query removes all the data from a table while retaining the table structure, resulting in improved performance compared to the DROP TABLE query.
- **DROP:** This DDL query deletes a database object, such as a table or an index, along with its associated data irreversibly.
- **RENAME:** The RENAME query changes the name of an existing database object, like altering the name of a table.

INSTRUCTIONS:

- As a part of LAB 3, there are 2 tasks that are to be completed wrt to the case study shared earlier with ER diagram and Relational Schema of University Fest Management.
 - **TASK 1:** As per the given Description, ER diagram and Relational Schema – create the tables using DDL commands and add the required constraints.
 - **TASK 2:** There are certain questions that have been given. These are to be executed on the DDL statements created in TASK 1.
- As a part of the submission process, the following are to be submitted:
 - A **PDF** document, containing all the Screenshots for both tasks as suggested
 - Name of the file: ``<your SRN>_University_Fest_DB_Lab3.pdf`
 - The **“.sql”** file for the same, shall contain all the commands that have been executed in the lab
 - Name of the file: ``<your SRN>_University_Fest_DB_Lab3.sql`

Example:

Refer to the sample submissions given below. This will give you an idea about the details that must be included in your submissions

NOTE: Screenshots can be taken either from “MySQL workbench” or “Command Line”

Task 1:

Take the screen shot of every table created along with the create table sql command.

Task 2:

Task 2 involves modifying the structure of the tables created in Task 1. For every modification made, **3 screenshots** are required:

1. Structure of the table **before** modification
2. **DDL statement**
3. Structure of the table **after** modification

Sample submission:

```
mysql> DESC student;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| SRN   | varchar(10)   | NO   | PRI | NULL    |       |
| name  | varchar(25)   | YES  |     | NULL    |       |
| section | char(1)       | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.18 sec)
```

```
mysql> ALTER TABLE student MODIFY section CHAR(1) DEFAULT 'A';
Query OK, 0 rows affected (0.26 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESC student;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| SRN   | varchar(10)   | NO   | PRI | NULL    |       |
| name  | varchar(25)   | YES  |     | NULL    |       |
| section | char(1)       | YES  |     | A       |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.01 sec)
```

TASK 1:

Identify all the constraints (domain, key, constraint on null, primary key, foreign key and check etc) based on the ER and description given and execute the DDL commands for University_fest database.

Add all the screenshots of sql command and desc table.

```
1 • CREATE DATABASE University_Fest;
2 • USE University_Fest;
3
4 • CREATE TABLE Fest (
5     Fest_ID INT PRIMARY KEY,
6     Fest_name VARCHAR(100) NOT NULL,
7     Year INT NOT NULL,
8     Head_teamID INT
9 );
10
11 • DESC Fest;
12
13 • CREATE TABLE Teams (
14     Team_ID INT PRIMARY KEY,
15     Team_name VARCHAR(100) NOT NULL,
16     Team_type VARCHAR(50),
17     FestID INT
18 );
19
20 • ALTER TABLE Fest
```

100% 11:11 1 error found

Result Grid Filter Rows: Search Export:

	Field	Type	Null	Key	Default	Extra
	Fest_ID	int	NO	PRI	NULL	
	Fest_name	varchar(100)	NO		NULL	
	Year	int	NO		NULL	
	Head_teamID	int	YES	MUL	NULL	

```

13 * CREATE TABLE Teams (
14     Team_ID INT PRIMARY KEY,
15     Team_name VARCHAR(100) NOT NULL,
16     Team_type VARCHAR(50),
17     FestID INT
18 );
19
20 * Desc Teams;
21
22 * ALTER TABLE Fest
23     ADD CONSTRAINT fk_fest_headteam
24     FOREIGN KEY (Head_teamID) REFERENCES Teams(Team_ID);

```

100% 12:20 1 error found

Result Grid



Filter Rows:



Search

Export:



	Field	Type	Null	Key	Default	Extra	
	Team_ID	int	NO	PRI	NULL		
	Team_name	varchar(100)	NO		NULL		
	Team_type	varchar(50)	YES		NULL		
	FestID	int	YES		NULL		

```

21
22 * ALTER TABLE Fest
23     ADD CONSTRAINT fk_fest_headteam
24     FOREIGN KEY (Head_teamID) REFERENCES Teams(Team_ID);
25
26 * ALTER TABLE Teams
27     ADD CONSTRAINT fk_team_fest
28     FOREIGN KEY (FestID) REFERENCES Fest(Fest_ID);
29
30

```

```

32 CREATE TABLE Members (
33     Mem_ID INT PRIMARY KEY,
34     Mem_name VARCHAR(100) NOT NULL,
35     DOB DATE,
36     super_MemID INT,
37     TeamID INT
38 );
39

```

```

40 ALTER TABLE Members ADD CONSTRAINT fk_members_id FOREIGN KEY (super_MemID) REFERENCES Members(Mem_ID);
41 ALTER TABLE Members ADD CONSTRAINT fk_team_id FOREIGN KEY (TeamID) REFERENCES Teams(Team_ID);
42

```

```

43 Desc Members;
44

```

```

45 CREATE TABLE Event (
46     Event_ID INT PRIMARY KEY,
47     Event_name VARCHAR(100) NOT NULL,

```

100% 14:43 1 error found

Result Grid Filter Rows: Search Export:

	Field	Type	Null	Key	Default	Extra	
	Mem_ID	int	NO	PRI	NULL		
	Mem_name	varchar(100)	NO		NULL		
	DOB	date	YES		NULL		
	super_MemID	int	YES	MUL	NULL		
	TeamID	int	YES		NULL		

```

45 CREATE TABLE Event (
46     Event_ID INT PRIMARY KEY,
47     Event_name VARCHAR(100) NOT NULL,
48     Building VARCHAR(50),
49     Floor INT,
50     Room_no VARCHAR(20),
51     Price DECIMAL(10,2),
52     TeamID INT
53 );
54 ALTER TABLE Event ADD CONSTRAINT fk_event_id FOREIGN KEY (TeamID) REFERENCES Teams(Team_ID);
55 Desc Event;
56

```

```

57 CREATE TABLE Event_conduction (
58     Event_ID INT,
59     Date_of_conduction DATE NOT NULL,

```

100% 12:55 1 error found

Result Grid Filter Rows: Search Export:

	Field	Type	Null	Key	Default	Extra	
	Event_ID	int	NO	PRI	NULL		
	Event_name	varchar(100)	NO		NULL		
	Building	varchar(50)	YES		NULL		
	Floor	int	YES		NULL		
	Room_no	varchar(20)	YES		NULL		
	Price	decimal(10,2)	YES		NULL		
	TeamID	int	YES	MUL	NULL		

```

66 CREATE TABLE Participants (
67     SRN VARCHAR(20) PRIMARY KEY,
68     name VARCHAR(100) NOT NULL,
69     department VARCHAR(50),
70     semester INT,
71     gender VARCHAR(10)
72 );

```

```

74 Desc Participants;

```

```

76 CREATE TABLE Visitors (
77     SRN VARCHAR(20),

```

100% 19:74 1 error found

Result Grid Filter Rows: Search Export:

	Field	Type	Null	Key	Default	Extra	
	SRN	varchar(20)	NO	PRI	NULL		
	name	varchar(100)	NO		NULL		
	department	varchar(50)	YES		NULL		
	semester	int	YES		NULL		
	gender	varchar(10)	YES		NULL		

```

57 • ⊖ CREATE TABLE Event_conduction (
58     Event_ID INT,
59     Date_of_conduction DATE NOT NULL,
60     PRIMARY KEY (Event_ID, Date_of_conduction),
61     FOREIGN KEY (Event_ID) REFERENCES Event(Event_ID)
62 );

```

```

64 • Desc Event_conduction;

```

```

66 • ⊖ CREATE TABLE Participants (
67     SRN VARCHAR(20) PRIMARY KEY,
68     name VARCHAR(100) NOT NULL,
69     department VARCHAR(50),
70     semester INT,

```

100% 23:64 1 error found

Result Grid



Filter Rows:



Search

Export:



	Field	Type	Null	Key	Default	Extra	
	Event_ID	int	NO	PRI	NULL		
	Date_of_conduction	date	NO	PRI	NULL		

```

76 • ⊖ CREATE TABLE Visitors (
77     SRN VARCHAR(20),
78     Name VARCHAR(100),
79     age INT,
80     gender VARCHAR(10),
81     PRIMARY KEY (SRN),
82     FOREIGN KEY (SRN) REFERENCES Participants(SRN)
83 );
84
85 • Desc Visitors;
86
87 • ⊖ CREATE TABLE Registration (
88     RegistrationID INT PRIMARY KEY,
89     Event_ID INT,

```

100% 1:75 1 error found

Result Grid



Filter Rows:



Search

Export:



	Field	Type	Null	Key	Default	Extra
	SRN	varchar(20)	NO	PRI	NULL	
	name	varchar(100)	NO		NULL	
	department	varchar(50)	YES		NULL	
	semester	int	YES		NULL	
	gender	varchar(10)	YES		NULL	


```

87 CREATE TABLE Registration (
88     RegistrationID INT PRIMARY KEY,
89     Event_ID INT,
90     SRN VARCHAR(20),
91     FOREIGN KEY (Event_ID) REFERENCES Event(Event_ID),
92     FOREIGN KEY (SRN) REFERENCES Participants(SRN)
93 );

```

```

95 Desc Registration;

```

```

97 CREATE TABLE Stall (
98     Stall_ID INT PRIMARY KEY,
99     name VARCHAR(100),
100    Fest_ID INT,

```

100% 19:95 1 error found

Result Grid



Filter Rows:

Search

Export:



	Field	Type	Null	Key	Default	Extra	
	RegistrationID	int	NO	PRI	NULL		
	Event_ID	int	YES	MUL	NULL		
	SRN	varchar(20)	YES	MUL	NULL		

```

97 CREATE TABLE Stall (
98     Stall_ID INT PRIMARY KEY,
99     name VARCHAR(100),
100     Fest_ID INT,
101     FOREIGN KEY (Fest_ID) REFERENCES Fest(Fest_ID)
102 );

```

```

104 Desc Stall;

```

100% 12:104 1 error found

Result Grid



Filter Rows:



Search

Export:



Field	Type	Null	Key	Default	Extra
Stall_ID	int	NO	PRI	NULL	
name	varchar(100)	YES		NULL	
Fest_ID	int	YES	MUL	NULL	

```

113 * ○ CREATE TABLE Stall_items (
114     Stall_ID INT,
115     Item_name VARCHAR(100),
116     Price_per_unit DECIMAL(10,2),
117     Total_quantity INT,
118     PRIMARY KEY (Stall_ID, Item_name),
119     FOREIGN KEY (Stall_ID) REFERENCES Stall(Stall_ID),
120     FOREIGN KEY (Item_name) REFERENCES Item(Name)
121 );
122
123
124 * DESC Stall_items;

```

100% 18:124

Result Grid



Filter Rows:

Search

Export:



	Field	Type	Null	Key	Default	Extra	
	Stall_ID	int	NO	PRI	NULL		
	Item_name	varchar(100)	NO	PRI	NULL		
	Price_per_unit	decimal(10,2)	YES		NULL		
	Total_quantity	int	YES		NULL		

```

106 CREATE TABLE Item (
107     Name VARCHAR(100) PRIMARY KEY,
108     Type VARCHAR(50)
109 );
110
111 Desc Item;

```

100% 11:111 1 error found

Result Grid Filter Rows: Search Export:

Field	Type	Null	Key	Default	Extra
Name	varchar(100)	NO	PRI	NULL	
Type	varchar(50)	YES		NULL	

```

120 CREATE TABLE Purchased(
121     SRN INT,
122     Stall_ID INT,
123     Item_Name VARCHAR(100),
124     Timestamp DATETIME,
125     Quantity INT,
126     PRIMARY KEY (SRN, Stall_ID, Item_Name, Timestamp),
127     FOREIGN KEY (SRN) REFERENCES Participants(SRN),
128     FOREIGN KEY (Stall_ID) REFERENCES Stall(Stall_ID),
129     FOREIGN KEY (Item_Name) REFERENCES Item(Name)
130 );
131
132 Desc Purchased;
133
134

```

100% 16:132

Result Grid Filter Rows: Search Export:

Field	Type	Null	Key	Default	Extra
SRN	varchar(20)	NO	PRI	NULL	
Stall_ID	int	NO	PRI	NULL	
Item_name	varchar(100)	NO	PRI	NULL	
Timestamp	timestamp	NO	PRI	CURRENT_TIMESTAMP	DEFAULT_GENERATED
Quantity	int	YES		NULL	

TASK 2:

1. Modify the datatype of the gender attribute to make sure that the only values that can be stored are M: for male, F: for female, and O: for other. Also, make sure that the gender attribute is positioned after the "name" column.

BEFORE

```
66 CREATE TABLE Participants (  
67     SRN VARCHAR(20) PRIMARY KEY,  
68     name VARCHAR(100) NOT NULL,  
69     department VARCHAR(50),  
70     semester INT,  
71     gender VARCHAR(10)  
72 );
```

```
74 Desc Participants;
```

```
76 CREATE TABLE Visitors (  
77     SRN VARCHAR(20),
```

100% 19:74 1 error found

Result Grid



Filter Rows:



Search

Export:



	Field	Type	Null	Key	Default	Extra	
	SRN	varchar(20)	NO	PRI	NULL		
	name	varchar(100)	NO		NULL		
	department	varchar(50)	YES		NULL		
	semester	int	YES		NULL		
	gender	varchar(10)	YES		NULL		

```

76 • CREATE TABLE Visitors (
77     SRN VARCHAR(20),
78     Name VARCHAR(100),
79     age INT,
80     gender VARCHAR(10),
81     PRIMARY KEY (SRN),
82     FOREIGN KEY (SRN) REFERENCES Participants(SRN)
83 );
84
85 • Desc Visitors;
86
87 • CREATE TABLE Registration (
88     RegistrationID INT PRIMARY KEY,
89     Event_ID INT,

```

100% 1:75 1 error found

Result Grid Filter Rows: Search Export:

	Field	Type	Null	Key	Default	Extra
	SRN	varchar(20)	NO	PRI	NULL	
	name	varchar(100)	NO		NULL	
	department	varchar(50)	YES		NULL	
	semester	int	YES		NULL	
	gender	varchar(10)	YES		NULL	

Command

ALTER TABLE Participants MODIFY gender ENUM('M', 'F', 'O') AFTER Name;

ALTER TABLE Visitors MODIFY gender ENUM('M', 'F', 'O') AFTER Name;

Desc Participants;

Desc Visitors;

AFTER

126 • ALTER TABLE Participants MODIFY gender ENUM('M', 'F', 'O') AFTER Name;

127 • ALTER TABLE Visitors MODIFY gender ENUM('M', 'F', 'O') AFTER Name;

128

129 • Desc Participants;

130 • Desc Visitors;

131

100% 15:130

Result Grid Filter Rows: Search Export:

	Field	Type	Null	Key	Default	Extra	
	SRN	varchar(20)	NO	PRI	NULL		
	name	varchar(100)	NO		NULL		
	gender	enum('M','F','O')	YES		NULL		
	department	varchar(50)	YES		NULL		
	semester	int	YES		NULL		

126 • ALTER TABLE Participants MODIFY gender ENUM('M', 'F', 'O') AFTER Name;

127 • ALTER TABLE Visitors MODIFY gender ENUM('M', 'F', 'O') AFTER Name;

128

129 • Desc Participants;

130 • Desc Visitors;

100% 15:130

Result Grid Filter Rows: Search Export:

	Field	Type	Null	Key	Default	Extra	
	SRN	varchar(20)	NO	PRI	NULL		
	Name	varchar(100)	YES		NULL		
	gender	enum('M','F','O')	YES		NULL		
	age	int	YES		NULL		

2. Every stall would offer items for different prices, it is found in the previous tests that the price of most of the items was 50. Therefore as a DB designer, set the default value of the prices of items to be 50 rupees and also make sure that every item has a price associated with it meaning we cannot have a null value entered into the price.

BEFORE

```
113 CREATE TABLE Stall_items (  
114     Stall_ID INT,  
115     Item_name VARCHAR(100),  
116     Price_per_unit DECIMAL(10,2),  
117     Total_quantity INT,  
118     PRIMARY KEY (Stall_ID, Item_name),  
119     FOREIGN KEY (Stall_ID) REFERENCES Stall(Stall_ID),  
120     FOREIGN KEY (Item_name) REFERENCES Item(Name)  
121 );  
122  
123  
124 DESC Stall_items;
```

100% 18:124

Result Grid Filter Rows: Search Export:

	Field	Type	Null	Key	Default	Extra
	Stall_ID	int	NO	PRI	NULL	
	Item_name	varchar(100)	NO	PRI	NULL	
	Price_per_unit	decimal(10,2)	YES		NULL	
	Total_quantity	int	YES		NULL	

Command

```
ALTER TABLE Stall_Items MODIFY Price_per_unit DECIMAL(10,2) NOT NULL DEFAULT 50;  
Desc Stall_Items;
```


AFTER

```
132 -- 2.
133 ALTER TABLE Stall_Items MODIFY Price_per_unit DECIMAL(10,2) NOT NULL DEFAULT 50;
134 DESC Stall_Items;
135
136
```

100% 18:134

Result Grid Filter Rows: Search Export:

	Field	Type	Null	Key	Default	Extra	
	Stall_ID	int	NO	PRI	NULL		
	Item_name	varchar(100)	NO	PRI	NULL		
	Price_per_unit	decimal(10,2)	NO		50.00		
	Total_quantity	int	YES		NULL		

3. Considering that the stalls have a limited space for storing the various items they sell, create a max_stocks condition that ensures that a particular stall can at max have 150 units of each item that they sell.

BEFORE

```
113 CREATE TABLE Stall_items (
114     Stall_ID INT,
115     Item_name VARCHAR(100),
116     Price_per_unit DECIMAL(10,2),
117     Total_quantity INT,
118     PRIMARY KEY (Stall_ID, Item_name),
119     FOREIGN KEY (Stall_ID) REFERENCES Stall(Stall_ID),
120     FOREIGN KEY (Item_name) REFERENCES Item(Name)
121 );
122
123
124 DESC Stall_items;
```

100% 18:124

Result Grid Filter Rows: Search Export:

	Field	Type	Null	Key	Default	Extra	
	Stall_ID	int	NO	PRI	NULL		
	Item_name	varchar(100)	NO	PRI	NULL		
	Price_per_unit	decimal(10,2)	YES		NULL		
	Total_quantity	int	YES		NULL		

Command

```
ALTER TABLE Stall_Items ADD CONSTRAINT chk_max_stock CHECK (Total_quantity <= 150);  
Desc Stall_Items;
```

AFTER

```
136 -- 3.  
137 ALTER TABLE Stall_Items ADD CONSTRAINT chk_max_stock CHECK (Total_quantity <= 150);  
138 Desc Stall_Items;
```

100% 18:138

Result Grid Filter Rows: Search Export:

	Field	Type	Null	Key	Default	Extra
	Stall_ID	int	NO	PRI	NULL	
	Item_name	varchar(100)	NO	PRI	NULL	
	Price_per_unit	decimal(10,2)	NO		50.00	
	Total_quantity	int	YES		NULL	

4: Rename the table "Event_conduction" to "Event_schedule"

BEFORE

```
57 CREATE TABLE Event_conduction (  
58     Event_ID INT,  
59     Date_of_conduction DATE NOT NULL,  
60     PRIMARY KEY (Event_ID, Date_of_conduction),  
61     FOREIGN KEY (Event_ID) REFERENCES Event(Event_ID)  
62 );  
63  
64 Desc Event_conduction;  
65  
66 CREATE TABLE Participants (  
67     SRN VARCHAR(20) PRIMARY KEY,  
68     name VARCHAR(100) NOT NULL,  
69     department VARCHAR(50),  
70     semester INT,
```

100% 23:64 1 error found

Result Grid Filter Rows: Search Export:

	Field	Type	Null	Key	Default	Extra
	Event_ID	int	NO	PRI	NULL	
	Date_of_conduction	date	NO	PRI	NULL	

Command

```
RENAME TABLE Event_conduction TO Event_schedule;
```

Desc Event_schedule;

AFTER

```
140      -- 4.
141      RENAME TABLE Event_conduction TO Event_schedule;
142      Desc Event_schedule;
143
144
145
```

100% 21:42

Result Grid

Filter Rows:

Search

Export: [illegible]

Result 17

Action Output

	Time	Action
✓ 43	15:12:38	ALTER TABLE Stall_Items ADD CONSTRAINT chk_max_stock CHECK (Total_quantity <= 150)
✓ 44	15:12:40	Desc Stall_Items
✗ 45	15:13:33	RENAME TABLE "Event_conduction" TO "Event_schedule"
✗ 46	15:14:38	RENAME TABLE "Event_conduction" TO "Event_schedule"
✓ 47	15:14:58	RENAME TABLE Event_conduction TO Event_schedule
✓ 48	15:15:10	Desc Event_schedule

5: Move the column "Date_of_conduction" such that it's the first column.

BEFORE

```

57 • CREATE TABLE Event_conduction (
58     Event_ID INT,
59     Date_of_conduction DATE NOT NULL,
60     PRIMARY KEY (Event_ID, Date_of_conduction),
61     FOREIGN KEY (Event_ID) REFERENCES Event(Event_ID)
62 );
63
64 • Desc Event_conduction;
65
66 • CREATE TABLE Participants (
67     SRN VARCHAR(20) PRIMARY KEY,
68     name VARCHAR(100) NOT NULL,
69     department VARCHAR(50),
70     semester INT,

```

100% 23:64 1 error found

Result Grid Filter Rows: Search Export:

	Field	Type	Null	Key	Default	Extra
	Event_ID	int	NO	PRI	NULL	
	Date_of_conduction	date	NO	PRI	NULL	

Command

ALTER TABLE Event_schedule MODIFY Date_of_conduction DATE FIRST;

Desc Event_schedule;

AFTER

```

144  -- 5.
145  * ALTER TABLE Event_schedule MODIFY Date_of_conduction DATE FIRST;
146  * Desc Event_schedule;
147
148
149

```

100% 21:146

Result Grid Filter Rows: Search Export:

Field	Type	Null	Key	Default	Extra
Date_of_conduction	date	NO	PRI	NULL	
Event_ID	int	NO	PRI	NULL	

Task 3: Few questions for you to answer:

1: Which is the sql command to know the current database in MySQL?

SELECT DATABASE(); shows the current database.

2: Which is the sql command to clear the command prompt window of MySQL?

system clear.

3: Can you rename the database in MySQL?

No.

4: What is the command to remove a table along with its structure?

DROP TABLE tablename; removes the table with structure.

5: Specify the difference between drop table and truncate table?

DROP deletes table and structure, TRUNCATE deletes only data but keeps structure.

6: Can a table have more than one primary key?

No, a table can have only one primary key, but can have multiple attributes.

7: Can a foreign key value be null?

Yes.

8: Can a primary key value be null? Which constraint is this?

No, the primary key cannot be NULL, enforced by the NOT NULL constraint.

9: Upon describing the table using the command “desc tablename” what information about the table is given.

Describe tablename; shows columns, types, keys, nullability, defaults.

10: Can a primary key for a table be changed? If yes how?

Yes, drop the old primary key and add a new one using ALTER TABLE.